

Aaron Gabayoyo

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EDUCATION

University of Edinburgh

Master of Informatics (MInf)

Edinburgh, Scotland

Sept 2022 – May 2027

SKILLS

Languages: C/C++, Java, Python, Scala

Compilers: LLVM, MLIR, TableGen, xDSL, ScaIR, SSA, Lit

Systems: RISC-V, Armv8, LLJIT, OpenMP, ThreadSanitizer

Tools: Git, CMake, CTest, Mill, Docker, GitHub Actions, Pytest, Ruff, Mypy

EXPERIENCE

Research Intern

July 2026 – Aug 2026

MLSystems CDT, University of Edinburgh

Edinburgh, Scotland

- Built *mlir-mracle*, a metamorphic testing harness that compares concurrent MLIR programs against semantics-preserving transformations to expose OpenMP bugs and miscompilations
- Fuzzed concurrent IR on 30+ litmus-style seed programs with 20+ Armv8 memory-model-aware transformations applied by a single MLIR pass
- Separated real failures from thread nondeterminism with a statistical oracle and ThreadSanitizer triage, sweeping LLJIT optimisation levels and OpenMP team sizes to expose divergent schedules
- **Discovered and reported a verifier gap in the MLIR OpenMP dialect** that permitted illegally nested `omp.critical` regions to compile; confirmed as a real bug and fixed upstream

Undergraduate Researcher — ScaIR Compiler Framework

Nov 2025 – Present

University of Edinburgh

Edinburgh, Scotland

- Contributed the interpreter component of ScaIR: an MLIR-compatible IR interpreter in Scala that executes MLIR SSA programs without lowering
- Benchmarked against xDSL over 10 runs at 1M-iteration scale, measuring **12–18x faster** interpretation on function-call, memref and Fibonacci programs
- Dispatched operations through a typed registry without downcasting, matching or beating xDSL's LoC in every dialect (up to 28.8% fewer) while adding compile-time type safety
- Implemented 25+ operations across 5 MLIR dialects (`arith`, `func`, `memref`, `scf` and LLVM terminators) supporting multi-block control flow, nested regions, loop-carried values and SSA-value scoping

PROJECTS

mlir-match | MLIR, C++, TableGen

Sept 2026

- Built an out-of-tree MLIR dialect for functional pattern matching, compiling patterns to Maranget-style decision trees to cut **comparisons per match by 2–7x** and match runtime by up to 3.04x
- Implemented the decision-tree pass by selecting each test column through a mixture heuristic, preferring the column that copies the fewest rows and then the one that splits them into the most groups
- Wrote 3 conversion passes lowering the dialect to `scf` and LLVM, verified through a lit suite

RISC-E | C++, RISC-V

July 2026 – Sept 2026

- Built a C++ tool for prototyping RV32I microarchitectures in software rather than RTL, swapping branch predictors, pipeline stalls and L1/L2 write-back caches to compare designs by simulated cost
- Wrote its RISC-V ELF loader, interpreter, syscall emulation and trap-based memory model, so each design executes real compiled binaries

BiggerBrother | Python, OpenCV, MediaPipe

June 2026 – July 2026

- Built computer-vision software for gym performance, able to derive rep metrics such as range of motion, tempo and concentric and eccentric splits, with load-velocity curves estimating a user's proximity to failure
- **Gated every push on CI with 120+ tests** (unit, integration, end-to-end) under pytest, plus ruff and mypy checks

ChocoPy Compiler | Python | Coursework

Jan 2025 – Apr 2025

- Wrote a compiler for ChocoPy, a statically typed subset of Python 3, including a parser, type-checker, optimisation passes and a lowering pass to RISC-SSA